THE LAB — DAY 03

The Pattern Finder

Finding hidden patterns — with no labels at all

Day 03 · Week 1 **Unlocks 🎨 Pattern Finder**

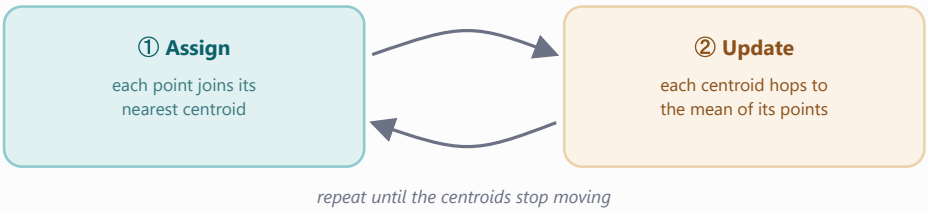
1 Mission briefing

A crate lands in the Lab with no label — thousands of measurements and **nobody** to say what they mean. No answer key, no teacher. Find the hidden structure yourself. *Previously:* you trained the Prediction Machine to name species from labeled examples. Today the labels are gone. This is **unsupervised learning**, and by the end of the day it powers your Studio's newest module.

- ☐ Tell **supervised** (answer key) from **unsupervised** (none) apart
- ☐ Run **k-means** to discover groups no one named
- ☐ Watch it **rediscover the penguin species** with zero labels
- ☐ Use the **elbow method** to guess how many groups exist
- ☐ Build a photo **color-compressor** with k-means
- ☐ Squash 4-D and 64-D data into pictures with **PCA** & **t-SNE**

2 Field guide the concepts

Unsupervised learning finds structure with no labels to copy. The workhorse is **k-means**. Picture three pizza shops opening in a new city with no map of customers. They play a two-move game, over and over: every customer orders from the **nearest** shop, then each shop moves to the **center** of the customers it just served. Repeat and the shops drift into the middle of three natural neighborhoods. Swap "shop" for **centroid** and "customer" for data point and that's the whole algorithm — *k* is how many shops you place.



k-means alternates **assign** ↔ **update** until it settles. Standardize features first so neither dominates.

Seeing high dimensions. A color is just a 3-D point (its R, G, B) — so k-means on a photo's pixels finds its palette and repaints it with a few colors (**compression**). And when data has 4 or 64 dimensions, **PCA** casts the most informative 2-D "shadow", while **t-SNE** untangles curvier structure into a map you can actually look at.

3 Lab missions check each off in the notebook

1 Let scikit-learn do the clustering ☐

`KMeans(n_clusters=3, n_init=10)` on two standardized measurements.

Expect: three clean blobs with red-X centroids.

2 The reveal: did it find the species? ☐

Cross-tabulate the clusters against the *true* species.

Expect: each cluster lines up mostly with one species — with zero labels.

3 Find the elbow ☐

Record `.inertia_` for k = 1…8 and plot it.

Expect: the curve drops steeply, then bends around k = 3.

4 Compress the photo to 8 colors ☐

Fit k-means on a pixel sample, then `.predict` every pixel.

Expect: the repaint still looks like the photo — in only 8 colors.

5 How few colors can you get away with? ☐

Repaint at k = 3, 8, 16 and compare.

Expect: k = 3 loses subtle shades; 8 vs 16 barely differ.

6 Flatten 4-D penguins onto a 2-D page ☐

`PCA(n_components=2)` on all four measurements.

Expect: the 2-D shadow keeps well over 80% of the information.

7 Draw the map ☐

Scatter the t-SNE embedding of 8×8 digit images, colored by digit.

Expect: roughly ten islands — one per digit, unprompted.

4 Cheat sheet clustering & dimensionality reduction

`KMeans(n_clusters=k, n_init=10).fit(X)`	Run assign/update until centroids settle
`km.fit_predict(X)`	Fit and return a cluster number per point
`km.labels_ · km.cluster_centers_`	Each point's cluster · the centroid positions
`km.inertia_`	Tightness of clusters (lower = tighter) — for the elbow
`km.predict(pixels)`	Label new points using a fitted model
`pd.crosstab(clusters, truth)`	Compare clusters to real labels in a grid
`StandardScaler().fit_transform(X)`	Rescale so no feature dominates the distance
`PCA(n_components=2).fit_transform(X)`	Best 2-D shadow; `.explained_variance_ratio_`
`TSNE(n_components=2, perplexity=30)`	Untangle high-D data into a 2-D map

5 🔓 Unlock your Studio module

🎨 Pattern Finder

Put your name on your work and pick a default number of colors; the unlock cell saves a small config token. It must write exactly:

`ai_studio/models/pattern_finder.json`

The clustering runs *live* in the app — upload any photo and it finds the dominant colors:

```
$ python ai_studio/app.py
```

6 Mini-glossary

unsupervised learning
Finding structure in data that has no labels to check against.

centroid
A cluster's center — the mean of the points assigned to it.

elbow method
Pick k where the inertia curve stops dropping sharply — its "bend".

t-SNE
A nonlinear map that pulls similar points together to reveal structure.

cluster
A group of points that resemble each other more than the rest.

inertia
How tightly points hug their centroid; smaller means tighter clusters.

PCA
Dimensionality reduction that keeps the most-informative straight-line "shadow".

7 Show & tell

1. k-means never saw a label — how well did its **3 clusters** match the real species? Show your crosstab.
2. Show your photo at **3, 8, and 16 colors**. Where's the sweet spot, and what falls apart at k = 3?
3. Your elbow bent at **k = 3**. If a mystery dataset bent at k = 5, what would that tell you?

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